

Experiment 2-No Obstacle: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1983	2.421	1.308	5.796	36.4%	69.0%	81.1%	13.919	28.399	9.047	23.115	0.572	1.304	1.440
Proposed	1972	2.391	1.043	4.079	47.4%	79.4%	87.1%	15.433	30.662	7.633	16.545	0.542	1.318	1.519
PPO-GRU	1988	2.815	1.279	8.661	38.4%	71.0%	83.4%	17.463	39.978	7.763	16.361	0.525	1.349	1.498
MPC	1109	6.032	3.090	16.630	2.6%	16.1%	46.7%	2.929	3.686	1.560	2.684	0.148	0.146	0.183

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1983	99.2%	17	0.8%	0.8%	17	100.0%	0	0
Proposed	2000	1972	98.6%	28	1.4%	1.4%	28	100.0%	0	0
PPO-GRU	2000	1988	99.4%	12	0.6%	0.6%	12	100.0%	0	0
MPC	2000	1109	55.5%	891	44.5%	44.5%	891	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	891	879	98.7%	206	23.4%	2.664
Proposed	891	871	97.8%	133	15.3%	2.642
PPO-GRU	891	881	98.9%	131	14.9%	2.691